Vector Calculus and Complex Analysis Homework I

1. Let f and g be scalar functions and $\boldsymbol{u}$ and $\boldsymbol{v}$ be vector functions having continuous first partial derivatives in some domain D in $\boldsymbol{R}^3$. Prove the followings:

(a) $\nabla(fg) = f\nabla g + g\nabla f$

(b) $\nabla(f/g) = (1/g^2)(g\nabla f - f\nabla g)$

(c) $\nabla^2(fg) = f\nabla^2 g + 2\nabla f \cdot \nabla g + g\nabla^2 f$

(d) $\operatorname{div}(f\boldsymbol{v}) = f\operatorname{div}(\boldsymbol{v}) + \boldsymbol{v}\cdot\nabla f$

(e) $\operatorname{curl}(f\boldsymbol{v}) = (\operatorname{grad} f)\times\boldsymbol{v} + f\operatorname{curl}\boldsymbol{v}$

(f) $\operatorname{div}(\boldsymbol{u}\times\boldsymbol{v}) = \boldsymbol{v}\cdot\operatorname{curl}\boldsymbol{u} - \boldsymbol{u}\cdot\operatorname{curl}\boldsymbol{v}$

Solution

(a) $$\nabla(fg) = \left(\frac{\partial(fg)}{\partial x}, \frac{\partial(fg)}{\partial y}, \frac{\partial(fg)}{\partial z}\right) = \left(f\frac{\partial g}{\partial x} + g\frac{\partial f}{\partial x}, f\frac{\partial g}{\partial y} + g\frac{\partial f}{\partial y}, f\frac{\partial g}{\partial z} + g\frac{\partial f}{\partial z}\right)$$
$$= \left(f\frac{\partial g}{\partial x}, f\frac{\partial g}{\partial y}, f\frac{\partial g}{\partial z}\right) + \left(g\frac{\partial f}{\partial x}, g\frac{\partial f}{\partial y}, g\frac{\partial f}{\partial z}\right) = \nabla(fg) = f\nabla g + g\nabla f$$

(b) $$\nabla(f/g) = \left(\frac{\partial(f/g)}{\partial x}, \frac{\partial(f/g)}{\partial y}, \frac{\partial(f/g)}{\partial z}\right) = (1/g^2)\left(g\frac{\partial f}{\partial x} - f\frac{\partial g}{\partial x}, g\frac{\partial f}{\partial y} - f\frac{\partial g}{\partial y}, g\frac{\partial f}{\partial z} - f\frac{\partial g}{\partial z}\right)$$
$$= (1/g^2)\left[\left(g\frac{\partial f}{\partial x}, g\frac{\partial f}{\partial y}, g\frac{\partial f}{\partial z}\right) - \left(f\frac{\partial g}{\partial x}, f\frac{\partial g}{\partial y}, f\frac{\partial g}{\partial z}\right)\right] = (1/g^2)(g\nabla f - f\nabla g)$$

(c) $$\nabla^2(fg) = \left(\frac{\partial^2(fg)}{\partial x^2}, \frac{\partial^2(fg)}{\partial y^2}, \frac{\partial^2(fg)}{\partial z^2}\right)$$
$$= \left(f\frac{\partial^2 g}{\partial x^2} + 2\frac{\partial f}{\partial x}\frac{\partial g}{\partial x} + g\frac{\partial^2 f}{\partial x^2}, f\frac{\partial^2 g}{\partial y^2} + 2\frac{\partial f}{\partial y}\frac{\partial g}{\partial y} + g\frac{\partial^2 f}{\partial y^2}, f\frac{\partial^2 g}{\partial z^2} + 2\frac{\partial f}{\partial z}\frac{\partial g}{\partial z} + g\frac{\partial^2 f}{\partial z^2}\right)$$
$$= \left(f\frac{\partial^2 g}{\partial x^2}, f\frac{\partial^2 g}{\partial y^2}, f\frac{\partial^2 g}{\partial z^2}\right) + \left(2\frac{\partial f}{\partial x}\frac{\partial g}{\partial x}, 2\frac{\partial f}{\partial y}\frac{\partial g}{\partial y}, 2\frac{\partial f}{\partial z}\frac{\partial g}{\partial z}\right) + \left(g\frac{\partial^2 f}{\partial x^2}, g\frac{\partial^2 f}{\partial y^2}, g\frac{\partial^2 f}{\partial z^2}\right)$$
$$= f\nabla^2 g + 2\nabla f \cdot \nabla g + g\nabla^2 f$$

(d) $\operatorname{div}(f\boldsymbol{v}) = \dfrac{\partial(fv_1)}{\partial x} + \dfrac{\partial(fv_2)}{\partial y} + \dfrac{\partial(fv_3)}{\partial z} = f\dfrac{\partial v_1}{\partial x} + v_1\dfrac{\partial f}{\partial x} + f\dfrac{\partial v_2}{\partial y} + v_2\dfrac{\partial f}{\partial y} + f\dfrac{\partial v_3}{\partial z} + v_3\dfrac{\partial f}{\partial z}$

$= f\dfrac{\partial v_1}{\partial x} + f\dfrac{\partial v_2}{\partial y} + f\dfrac{\partial v_3}{\partial z} + v_1\dfrac{\partial f}{\partial x} + v_2\dfrac{\partial f}{\partial y} + v_3\dfrac{\partial f}{\partial z} = f\operatorname{div}(\boldsymbol{v}) + \boldsymbol{v}\cdot\nabla f$

(e) $\operatorname{curl}(f\boldsymbol{v}) = \left(\dfrac{\partial(fv_3)}{\partial y} - \dfrac{\partial(fv_2)}{\partial z}\right)\mathbf{i} + \left(\dfrac{\partial(fv_1)}{\partial z} - \dfrac{\partial(fv_3)}{\partial x}\right)\mathbf{j} + \left(\dfrac{\partial(fv_2)}{\partial x} - \dfrac{\partial(fv_1)}{\partial y}\right)\mathbf{k}$

$= \left(f\dfrac{\partial v_3}{\partial y} + v_3\dfrac{\partial f}{\partial y} - f\dfrac{\partial v_2}{\partial z} - v_2\dfrac{\partial f}{\partial z}\right)\mathbf{i} + \left(f\dfrac{\partial v_1}{\partial z} + v_1\dfrac{\partial f}{\partial z} - f\dfrac{\partial v_3}{\partial x} - v_3\dfrac{\partial f}{\partial x}\right)\mathbf{j} + \left(f\dfrac{\partial v_2}{\partial x} + v_2\dfrac{\partial f}{\partial x} - f\dfrac{\partial v_1}{\partial y} - v_1\dfrac{\partial f}{\partial y}\right)\mathbf{k}$

$= \left(v_3\dfrac{\partial f}{\partial y} - v_2\dfrac{\partial f}{\partial z}\right)\mathbf{i} + \left(v_1\dfrac{\partial f}{\partial z} - v_3\dfrac{\partial f}{\partial x}\right)\mathbf{j} + \left(v_2\dfrac{\partial f}{\partial x} - v_1\dfrac{\partial f}{\partial y}\right)\mathbf{k}$

$+ f\left[\left(\dfrac{\partial v_3}{\partial y} - f\dfrac{\partial v_2}{\partial z}\right)\mathbf{i} + \left(\dfrac{\partial v_1}{\partial z} - \dfrac{\partial v_3}{\partial x}\right)\mathbf{j} + \left(\dfrac{\partial v_2}{\partial x} - \dfrac{\partial v_1}{\partial y}\right)\mathbf{k}\right]$

$= (\operatorname{grad} f)\times\boldsymbol{v} + f\operatorname{curl}\boldsymbol{v}$

(f) $\operatorname{div}(\boldsymbol{u}\times\boldsymbol{v}) = \dfrac{\partial(u_2v_3 - u_3v_2)}{\partial x} + \dfrac{\partial(u_3v_1 - u_1v_3)}{\partial y} + \dfrac{\partial(u_1v_2 - u_2v_1)}{\partial z}$

$= v_3\dfrac{\partial u_2}{\partial x} + u_2\dfrac{\partial v_3}{\partial x} + v_1\dfrac{\partial u_3}{\partial y} + u_3\dfrac{\partial v_1}{\partial y} + v_2\dfrac{\partial u_1}{\partial z} + u_1\dfrac{\partial v_2}{\partial z}$

$-u_3\dfrac{\partial v_2}{\partial x} - v_2\dfrac{\partial u_3}{\partial x} - u_1\dfrac{\partial v_3}{\partial y} - v_3\dfrac{\partial u_1}{\partial y} - u_2\dfrac{\partial v_1}{\partial z} - v_1\dfrac{\partial u_2}{\partial z}$

$= \left[v_1\left(\dfrac{\partial u_3}{\partial y} - \dfrac{\partial u_2}{\partial z}\right) + v_2\left(\dfrac{\partial u_1}{\partial z} - \dfrac{\partial u_3}{\partial x}\right) + v_3\left(\dfrac{\partial u_2}{\partial x} - \dfrac{\partial u_1}{\partial y}\right)\right]$

$-\left[u_1\left(\dfrac{\partial v_3}{\partial y} - \dfrac{\partial v_2}{\partial z}\right) + u_2\left(\dfrac{\partial v_1}{\partial z} - \dfrac{\partial v_3}{\partial x}\right) + u_3\left(\dfrac{\partial v_2}{\partial x} - \dfrac{\partial v_1}{\partial y}\right)\right]$

$= \boldsymbol{v}\cdot\operatorname{curl}\boldsymbol{u} - \boldsymbol{u}\cdot\operatorname{curl}\boldsymbol{v}$

2. Compute curl **F** for

(a) $\boldsymbol{F}(x, y, z) = (x^2y, 3x - yz, z^3)$,

(b) $\boldsymbol{F}(x, y, z) = (x^3 - y, y^5, e^z)$.

Solution

for (a):

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ x^2 y & 3x - yz & z^3 \end{vmatrix}$$

$$= \left(\frac{\partial(z^3)}{\partial y} - \frac{\partial(3x - yz)}{\partial z} \right) \mathbf{i} - \left(\frac{\partial(z^3)}{\partial x} - \frac{\partial(x^2 y)}{\partial z} \right) \mathbf{j} + \left(\frac{\partial(3x - yz)}{\partial x} - \frac{\partial(x^2 y)}{\partial y} \right) \mathbf{k}$$

$$= (0 + y)\mathbf{i} - (0 - 0)\mathbf{j} + (3 - x^2)\mathbf{k} = \langle y, 0, 3 - x^2 \rangle.$$

Similarly, for part (b), we have

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ x^3 - y & y^5 & e^z \end{vmatrix}$$

$$= \left(\frac{\partial(e^z)}{\partial y} - \frac{\partial(y^5)}{\partial z} \right) \mathbf{i} - \left(\frac{\partial(e^z)}{\partial x} - \frac{\partial(x^3 - y)}{\partial z} \right) \mathbf{j} + \left(\frac{\partial(y^5)}{\partial x} - \frac{\partial(x^3 - y)}{\partial y} \right) \mathbf{k}$$

$$= (0 - 0)\mathbf{i} - (0 - 0)\mathbf{j} + (0 + 1)\mathbf{k} = \langle 0, 0, 1 \rangle.$$

3. Calculate $\int_C \boldsymbol{F}(\boldsymbol{r}) \cdot d\boldsymbol{r}$ for

(a) $\boldsymbol{F}(x, y) = (2y + x^2, x^2 - 2x)$, C is the upper half-circle from $(-3, 0)$ to $(3, 0)$

(b) $\boldsymbol{F}(x, y) = (x^2 e^y, y e^x)$, C is the portion of $y = x^3$ from $(0, 0)$ to $(1, 1)$

(c) $\boldsymbol{F}(x, y, z) = (xy, 3z, 1)$, C is the helix $x = \cos t$, $y = \sin t$, $z = 2t$ from $(1, 0, 0)$ to $(0, 1, \pi)$

Solution

a. $x = -3\cos t, y = 3\sin t$

$0 \le t \le \pi$

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

$$= \int_C \left(2y + x^2\right) dx + \left(x^2 - 2x\right) dy$$

$$= \int_0^{\pi} \left[\left(2\left(3\sin t\right) + 9\cos^2 t\right) 3\sin t + \left(9\cos^2 t + 6\cos t\right) 3\cos t\right] dt$$

$$= \int_0^{\pi} \left[18 + 27\left(\sin t + \cos t\right)\cos^2 t\right] dt$$

$$= 18 + 18\pi$$

b. $x = t, y = t^3$

$0 < t < 1$

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

$$= \int_C x^2 e^y dx + y e^x dy$$

$$= \int_0^1 \left(t^2 e^{t^3} + \left(t^3 e^t\right) 3t^2\right) dt$$

$$= \int_0^1 t^2 e^{t^3} dt + 3\int_0^1 t^5 e^t dt$$

$$= \frac{1079}{3} - \frac{395}{3} e$$

c. $x = \cos t \quad y = \sin t \quad z = 2t$

$0 \le t \le \pi/2$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C xy\, dx + 3z\, dy + 1\, dz$$

$$= \int_0^{\pi/2} [(\cos t \sin t)(-\sin t) + 3(2t)(\cos t) + 1(2)]\, dt$$

$$= \int_0^{\pi/2} \left[-\cos t \sin^2 t + 6t \cos t + 2\right]\, dt$$

$$= -\frac{1}{3} + (3\pi - 6) + \pi = 4\pi - \frac{19}{3}$$

4. Calculate

(a) $\int_{(4,0)}^{(6,1)} e^{4y}(2xdx + 4x^2dy)$

(b) $\int_{(0,0,\pi)}^{(2,1/2,\pi/2)} e^{xy}(y\sin z dx + x\sin z dy + \cos z dz)$

Solution

(a) $e^{4y}(2xdx + 4x^2dy) = \nabla(x^2e^{4y}) \cdot (dx, dy)$

$\int_{(4,0)}^{(6,1)} e^{4y}(2xdx + 4x^2dy) = x^2e^{4y}\Big|_{(4,0)}^{(6,1)} = 36e^4 - 16$

(b) $e^{xy}(y\sin z dx + x\sin z dy + \cos z dz) = \nabla(e^{xy}\sin z) \cdot (dx, dy, dz)$

$\int_{(0,0,\pi)}^{(2,1/2,\pi/2)} e^{xy}(y\sin z dx + x\sin z dy + \cos z dz) = e^{xy}\sin z\Big|_{(0,0,\pi)}^{(2,1/2,\pi/2)} = e$